aie26

Topic 06 : Data Modelling

Part 02 : Column Encoding Scaling

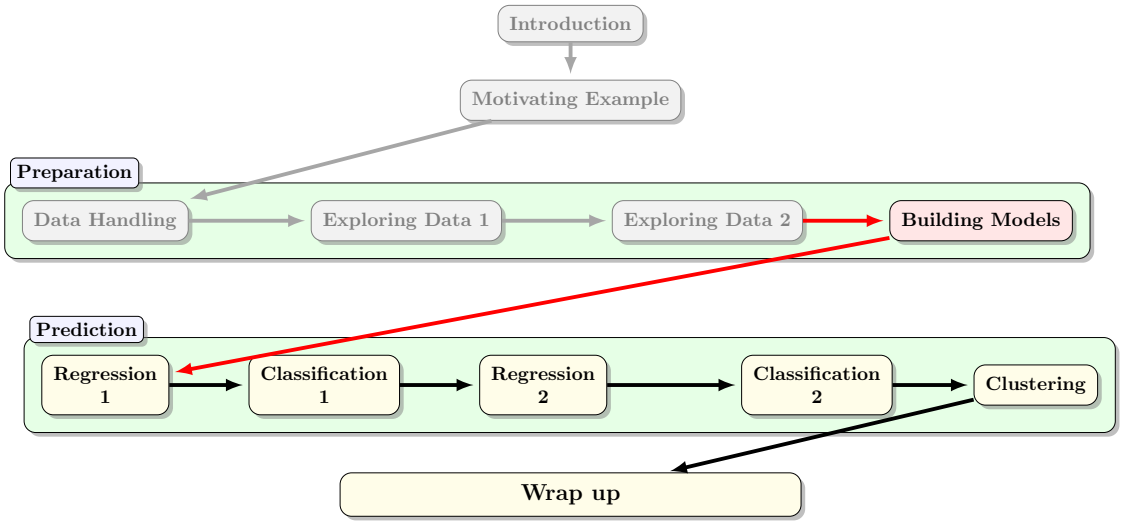
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Autumn Semester, 2026

Outline

- Encoding categorical features, using pandas
- Scaling numeric features
- Looking ahead to feature preparation
- Overview of ML and what we have achieved

AI Engineering (Week 6)



Outline

1. Feature engineering 3

2. Categorical Columns 6
2.1. Using Pandas 7
2.2. Using Scikit-learn 14

3. Wrap up 24

4. Resources 29

Featuring engineering 1: Scaling of numerical variables

Scaling - what it does

- If numeric features have different scales, e.g. [-0.005, -0.003] and [10000, 10001] some terms dominate, others are "lost"
- Better: transfer the scaling from the feature to the model parameter
- A min-max scaling is often a good choice:

$$\tilde{X} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

- Note that X is in the range $[X_{\min}, X_{\max}]$ but $\tilde{X}$ is in the range $[0, 1]$.
- Other options include StandardScaler (subtract mean and divide by standard deviation) and a max-abs scaler (scales to [-1,1])
- Whatever scaler is used, it should be trained using the training set only. It can be applied to either training or test set.

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- Whatever scaler is used, it should be trained using the training set only. It can be applied to either training or test set.

sklearn example

```
from sklearn.preprocessing import MinMaxScaler
# df is a dataframe with numeric features
scaler = MinMaxScaler()
dfScaled = scaler.fit(df))
```

`dfScaled` can be used instead of `df` with the advantage that the fitted parameters are more accurate.

Feature Engineering 2: Choice of Features

- How many to include? Use metrics to decide. Will see some when considering regression and classification.
- How do we handle different feature types? Need to encode categorical variables.
- Can we derive new numeric features? Yes, $f' = \log(f)$ etc. is possible

Scaling Summary

- Scaling is applied to *numerical features* only.
- Consistent choices should be made, so that all numerical features are given equal weight when fitting the training set. Be careful of outliers!
- Scaling should be applied consistently to test and unseen data, because feature scaling becomes part of the fitted model.

Type	Typical Range	When to use
`StandardScaler`	[-3, 3]	Relative distribution of features is important, e.g., some but not all features are skewed
`MaxAbsScaler`	[0,1], [-1,0], [-1, 1]	Feature distribution differences are less important, just need scaled features to share the same range
`MinMaxScaler`	[0,1]	As `MaxAbsScaler`, but also require all scaled features to be non-negative (e.g., (scaled) `income` should be nonnegative.

scikit-learn has many more scalers, quantizers and transformers for numerical data!

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Using Categorical Features in (Logistic) Regression

How can Categorical-valued features participate in linear models?

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How can Categorical-valued features participate in linear models?

Given the following fragment of a dataset, where the goal is to predict the salary of employees in a large organisation:

```
df = pd.read_csv('data/team.csv',\
       index_col="Name")
df
```

Name	Role	Skilled	Salary
Alice	Designer	Yes	40000
Bob	Programmer	No	25000
Carol	Tester	No	30000

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How can this data be represented by a linear model, where all quantities must take numeric values?

Using pandas .getdummies() on a binary-valued column

```
dfSkilledDummies = pd.get_dummies(df['Skilled'],\
       prefix='Skilled',\
       dtype=int)
dfSkilledDummies
```

Name	Skilled_No	Skilled_Yes
Alice	0	1
Bob	1	0
Carol	1	0

Using pandas .getdummies() on a binary-valued column

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dfSkilledDummies
```

Name	Skilled_No	Skilled_Yes
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Bob	1	0
Carol	1	0

Note that a binary-valued column becomes 2 dummy columns

Reducing redundancy (by 1) in 2 dummy columns

```
dfSkilledIndicators = pd.get_dummies(df['Skilled'],\
       prefix='Skilled',\
       drop_first=True,\
       dtype=int)\
       .rename(columns={"Skilled_Yes": "IsSkilled"})
dfSkilledIndicators
```

Name	IsSkilled
Alice	1
Bob	0
Carol	0

Reducing redundancy (by 1) in 2 dummy columns

```
dfSkilledIndicators = pd.get_dummies(df['Skilled'],\
       prefix='Skilled',\
       drop_first=True,\
       dtype=int)\
       .rename(columns={"Skilled_Yes": "IsSkilled"})
dfSkilledIndicators
```

Name	IsSkilled
Alice	1
Bob	0
Carol	0

A single indicator column can replace a group of 2 dummy columns

Using pandas .getdummies() on a multi-valued column

```
dfRoleDummies = pd.get_dummies(df['Role'],\
       prefix='Role',\
       dtype=int)
dfRoleDummies
```

Name	Role_Designer	Role_Programmer	Role_Tester
Alice	1	0	0
Bob	0	1	0
Carol	0	0	1

Using pandas .getdummies() on a multi-valued column

```
dfRoleDummies = pd.get_dummies(df['Role'],\
       prefix='Role',\
       dtype=int)
dfRoleDummies
```

Name	Role_Designer	Role_Programmer	Role_Tester
Alice	1	0	0
Bob	0	1	0
Carol	0	0	1

Note that an n-valued column becomes n dummy columns

Reducing redundancy (by 1) in *n* dummy columns

```
dfRoleIndicators = pd.get_dummies(df['Role'],\
       prefix='Role',\
       drop_first=True,\
       dtype=int)\
       .rename(columns={\
       "Role_Programmer": "IsProgrammer",\
       "Role_Tester": "IsTester"})
dfRoleIndicators
```

Name	IsProgrammer	IsTester
Alice	0	0
Bob	1	0
Carol	0	1

Reducing redundancy (by 1) in n dummy columns

```
dfRoleIndicators = pd.get_dummies(df['Role'],\
       prefix='Role',\
       drop_first=True,\
       dtype=int)\
       .rename(columns={\
       "Role_Programmer": "IsProgrammer",\
       "Role_Tester": "IsTester"})
dfRoleIndicators
```

Name	IsProgrammer	IsTester
Alice	0	0
Bob	1	0
Carol	0	1

$n - 1$ indicator columns can replace a group of n dummy columns

Deriving and using dummy/indicator features

- Identify potential categorical features in EDA Pass 1
- Identify whether each feature is (potentially) *usable* in EDA Pass 2
- Identify whether each feature is (potentially) *useful* in EDA Pass 3
- Add all potentially usable and useful features (regardless of type) to a list F
- For each categorical feature f_j in F having n levels
  - Derive $n-1$ indicator features $\tilde{f}_j^k$, where $k = 1, \ldots, n-1$
  - Replace the original categorical feature f_j in F with the derived indicator features $\tilde{f}_j^k$.
- Build the model using the features in F.

Analysis of pandas dummies

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3. How do we handle data (that includes categorical columns) that is split into training and test?

Analysis of pandas dummies

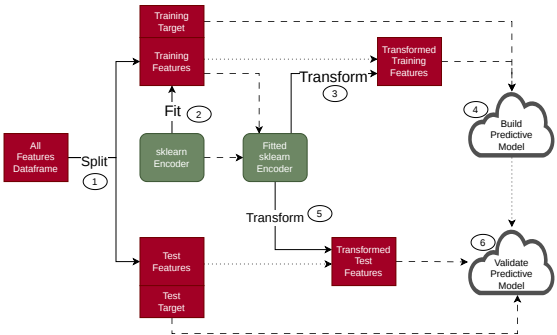
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3. How do we handle data (that includes categorical columns) that is split into training and test?
4. How can we reverse the operation (i.e., return from (0,1)-valued columns to categorical columns)?

Flow of operations with scikit-learn encoders

- The numbers indicate the flow of operations when scikit-learn's built-in Encoders are used with features.
- As with scaling, the Encoder is "trained" using `.fit()` on the training features only.
- The fitted encoder is applied to the training features using `.transform()` before they are used to build the model.
- The same procedure is applied to the test features so they can be used with the model.
- The transformed features *replace* the original features directly.
- scikit-learn Encoders also provide `.inverse_transform()` to restore the original features.



Worked example: Viewing the dataset

The dataset has a mixture of columns that need to be encoded

- Unordered feature category: `colour`, taking values `blue`, `green` and `red`.
- Ordered feature category: `size`, taking values `small`, `medium` and `large`.
- Date/time feature category: `hour`, taking values from 0 to 23.
- Labeled target: `move`, taking values `down`, `left`, `right` and `up`.

	hour	colour	size	move
32	8	green	medium	left
18	18	blue	small	left
17	17	blue	large	up
33	9	red	large	up
27	3	blue	large	right

Worked example: using `OneHotEncoder`

```
def applyOHEdirect(oheEnc, df, colName,\
                   categories):
  # Apply mapping
  ohe = oheEnc.transform(df[[colName]])
  colNames = ['is_' + x for x in categories]
  dfOHEsk = pd.DataFrame(ohe.toarray(),\
       columns=colNames,\
       index=df.index)
  return dfOHEsk
```

```
colName = 'colour'
# dummies are ints
oheEnc = skp.OneHotEncoder(dtype=int)
# "learn" mapping to dummies
oheEnc.fit(dfTrain[[colName]])
```

```
categories = oheEnc.categories_
dfOheColoursSk = applyOHEdirect(oheEnc,\
       dfTrain, colName, categories)
dfOheColoursSk.head()
```

- `fit` the `OneHotEncoder` to the training data, to learn the mapping from the training set
- `transform` the data (training data is used here, but it could also be applied to test data)
- Note that we try to make the transformed columns easier to interpret

Worked example: `OneHotEncoder` results and inverse

	is_blue	is_green	is_red
32	0	1	0
18	1	0	0
17	1	0	0
33	0	0	1
27	1	0	0

```
# Complete the round trip from unordered category
# to OHE and back
colourOheArr = dfOheColoursSk.values
dfColourRoundTrip = pd.DataFrame(\
    oheEnc.inverse_transform(dfOheColoursSk),\
    index=dfTrain.index,\
    columns=[colName])
dfColourRoundTrip.head()
```

	colour
32	green
18	blue
17	blue
33	red
27	blue

- The transformed OHE colour columns have 0,1 values instead of colour names
- The `inverse_transform` restores the original (`blue`, `green`, `red`) labels (compare with the original training data on earlier slide)

Worked example: using `OrdinalEncoder`

```
# Derive integers in the right order for
# the ordinal category
colName = 'size'
categories=[['small','medium','large']]
ordinalEnc = skp.OrdinalEncoder(\
    categories=categories, dtype=int)
sizeArr = dfTrain[colName]\
    .values.reshape(-1,1)
ordinalEnc.fit(sizeArr)
```

```
ordName = 'ord_' + colName
dfSizeOrdinal = pd.DataFrame(\
    ordinalEnc.transform(sizeArr),\
    index=dfTrain.index,\
    columns=[ordName])
dfSizeOrdinal.head()
```

- `fit` the `OrdinalEncoder` to `size` in the training data, to learn the mapping from the training set
- As before, `transform` the data (training data is used here, but it could also be applied to test data)
- We distinguish between the transformed columns and the original `size` column

Worked example: `OrdinalEncoder` results and inverse

	ord_size
32	1
18	0
17	2
33	2
27	2

```
# Complete the round trip from ordinal to
# integer and back
sizeOrdinalArr = dfSizeOrdinal.values
dfSizeOrdinalRoundTrip = pd.DataFrame(\
    ordinalEnc.inverse_transform(\
        sizeOrdinalArr),\
    index=dfTrain.index,\
    columns=[colName])
dfSizeOrdinalRoundTrip.head()
```

	size
32	medium
18	small
17	large
33	large
27	large

- The transformed Ordinal `size` columns have 0,1,2 values instead of size labels
- The `inverse_transform` restores the original (`small`, `medium`, `large`) labels (compare with the original training data on earlier slide)

Worked example: using `LabelEncoder`

```
# Apply label encoder to the target.
# NB: this encoder is _intended for target
# columns only_.
# DO NOT USE label encoder on features!!!!
colName = 'move'
labelEnc = skp.LabelEncoder()
y = dfTrain[colName]
# See https://stackoverflow.com/a/36120015/1988855
labelEnc.fit(y)
```

```
lblName = 'lbl_' + colName
dfMoveLbl = pd.DataFrame(\
    labelEnc.transform(y),\
    index=dfTrain.index,\
    columns=[lblName])
dfMoveLbl.head()
```

- The `LabelEncoder` should be used with target columns only.
- Never use `LabelEncoder` with feature columns!!!

Worked example: `LabelEncoder` results and inverse

	lbl_move
32	1
18	1
17	3
33	3
27	2

```
# Complete the round trip from label
# to integer and back
yE = dfMoveLbl['lbl_move']
dfMoveLblRoundTrip = pd.DataFrame(\
    labelEnc.inverse_transform(yE),\
    index=dfTrain.index,\
    columns=[colName])
dfMoveLblRoundTrip.head()
```

	move
32	left
18	left
17	up
33	up
27	right

- The Label-encoded transformed target (`move`) columns have 0,1,2,3 values instead of `move` labels
- The `inverse_transform` restores the original (`down`, `left`, `right`, `up`) labels (compare with the original training data on earlier slide)

Bonus 1: manual `CyclicEncoder`

```
def cyclicEncode(inDf, colName, period):
  encoded = {
  colName+'_cos' : np.cos(2 * np.pi * inDf[colName]/period),
  colName+'_sin' : np.sin(2 * np.pi * inDf[colName]/period)
  }
  outDf = pd.DataFrame(encoded, index=inDf.index)
  return outDf
```

```
encodedHour = cyclicEncode(dfTrain, 'hour', 24)
encodedHour.head()
```

	hour_cos	hour_sin
32	-5.000000e-01	0.866025
18	-1.836970e-16	-1.000000
17	-2.588190e-01	-0.965926
33	-7.071068e-01	0.707107
27	7.071068e-01	0.707107

- Cyclic encoding is intended for quantities that “wrap around”, like hours, months, etc.
- The formula is well-defined and its parameters do not need to be learnt from the training data
- It can be applied directly as shown, but would need to be adjusted to have the same API as a sklearn equivalent encoder.

Bonus 2: manual `ReverseCyclicEncoder`

```
def reverseCyclicEncode(inDf, colName, period):
  c = inDf[colName+'_cos'].to_numpy()
  s = inDf[colName+'_sin'].to_numpy()
  theta = np.atan2(s,c); twopi = 2.0*np.pi
  # Ensure theta angle is between 0 and 2*pi
  thetaNN = np.where(theta < 0, theta+twopi, theta)
  decoded = {
    colName+'_inv' : (thetaNN * period / twopi)
  }
  outDf = pd.DataFrame(decoded, index=inDf.index)
  return outDf
```

```
decodedHour = reverseCyclicEncode(encodedHour, 'hour', 24)
decodedHour.head()
```

	hour_inv
32	8.0
18	18.0
17	17.0
33	9.0
27	3.0

- Reverse cyclic encoding inverts the cyclic encoding, back to the original variable (`hour`) numbers.
- It also does not need to be “learnt” from data, just applied as an expression.
- Unlike OHE, Ordinal and Label encoding, the cyclic encoding round trip does not return exactly the same values, due to roundoff error when computing trigonometric functions.

Encoding Summary

- Any column (feature or target) can be *encoded* so that it can participate in a model.
- Commonly used to encode categorical features and targets, but other uses include cyclic encoding and feature contrasts
- Test and unseen data should also be encoded, because encoding becomes part of the fitted model.

Type	Effect	When to use
Pandas dummies	Unordered Categorical feature with l levels $\rightarrow$ l or $l-1$ (0,1) indicator features	Transform categorical feature to numerical, but inverse transformation is not needed.
Sklearn `OneHotEncoder`	Unordered Categorical feature with l levels $\rightarrow$ l or $l-1$ (0,1) indicator features	Transform categorical feature to numerical. Inverse transformation is needed.
Sklearn `OrdinalEncoder`	Ordered Categorical feature with l levels $\rightarrow$ (0,1,...,l) feature	Transform ordered categorical feature to numerical.
Sklearn `LabelEncoder`	Categorical target with l levels $\rightarrow$ (0,1,...,l) target	Transform categorical target to numerical.

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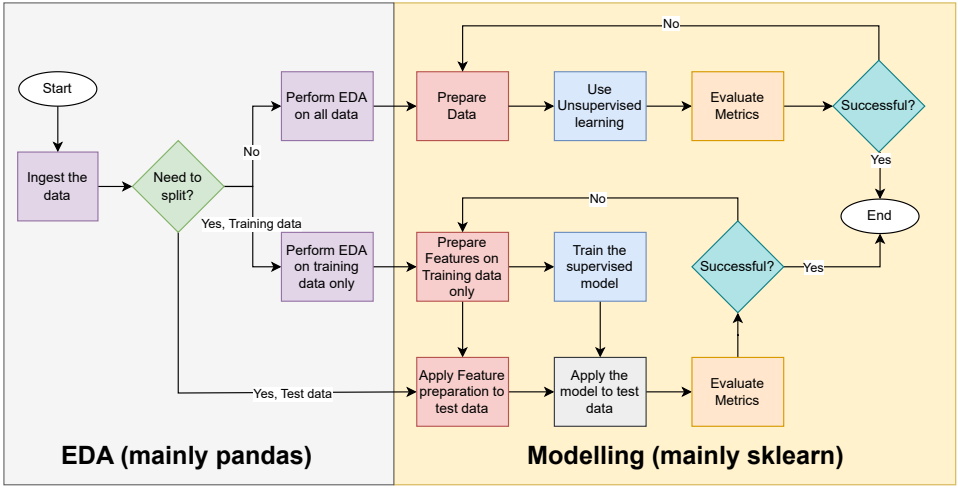
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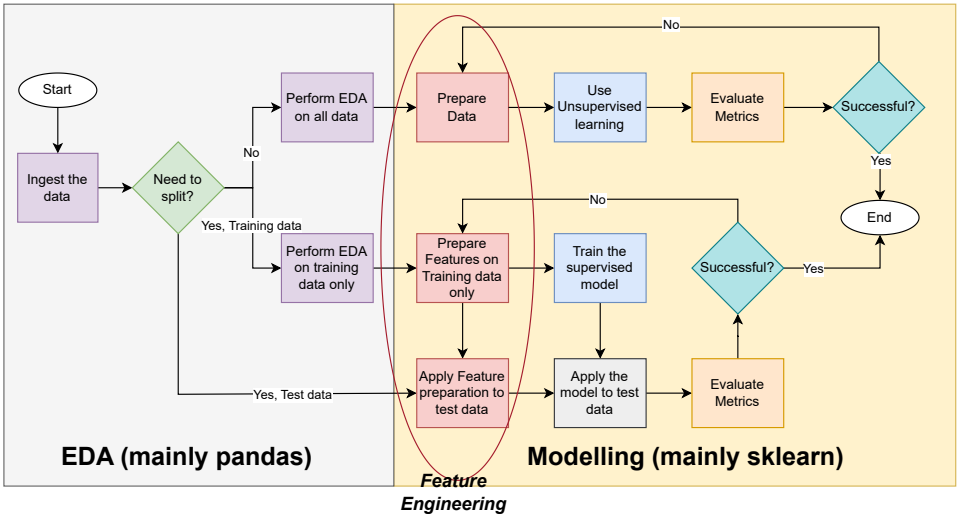
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An overview of Machine Learning



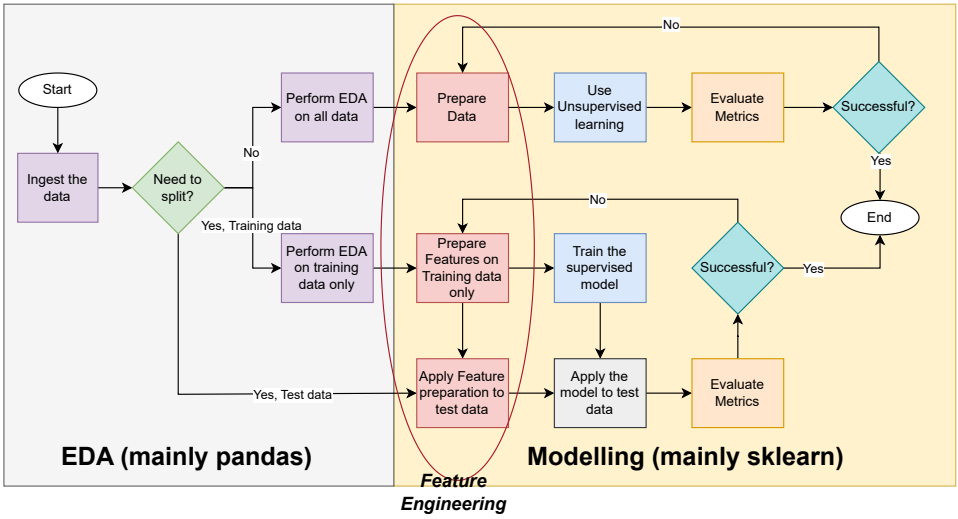
On first glance, this might seem overwhelming, but note that the boxes are colour-coded, so that related operations share the same colour.

Where feature preparation fits. . .



NB: Feature preparation is informed by EDA *but is not part of EDA*.

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Feature Engineering more generally

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- Use of pandas `getdummies()` requires care and is not always suitable
- Scikit-learn, imported as the `sklearn` package, supports a variety of column transformations
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- Can also rescale numerical columns, or encode more exotic columns (other datatypes, computed columns, . . .)

Feature Engineering more generally

- Use of pandas `getdummies()` requires care and is not always suitable
- Scikit-learn, imported as the `sklearn` package, supports a variety of column transformations
- Categorical columns can be features or targets, ordered or unordered
- Can also rescale numerical columns, or encode more exotic columns (other datatypes, computed columns, . . .)
- As seen in the schematic, if an ML procedure is unsuccessful, more feature engineering should be considered - it can help.

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  - Encoding categorical features as numerical dummy features (more to come in future weeks…), so they can participate in linear models
- In subsequent weeks we will put this theory into practice.

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Resources

- **A Summary of the Basic Machine Learning Models**
  towardsdatascience.com/a-summary-of-the-basic-machine-learning-models-e0a65627ecbe

- **Train-Test Split for Evaluating Machine Learning Algorithms**
  https://machinelearningmastery.com/
  train-test-split-for-evaluating-machine-learning-algorithms
  This week I have focused on the theory rather than its (python) implementation. This is a nice article that covers the implementation side of things.

- **Cross-Validation: Estimator Evaluator**
  medium.com/swlh/cross-validation-estimator-evaluator-897d28afb4ff
  Nice article that covers cross-validation in a lot more detail — we will be using many of these variants in later weeks, especially k-fold stratified.